

Many-Worlds Is the Simplest Interpretation: An Algorithmic Argument

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Abstract

The Many-Worlds Interpretation adds nothing to the Schrödinger equation. Collapse interpretations add non-unitary processes; hidden-variable theories add unobservable particles and pilot waves. I argue this difference in descriptive complexity has ontological consequences. Under algorithmic probability—the Solomonoff prior extended from epistemology to ontology—interpretations with lower Kolmogorov complexity dominate the measure over physical theories. Many-Worlds, requiring only unitary evolution, has minimal K and is therefore most probable. This transforms “MWI is simpler” from aesthetic preference to measure-theoretic claim. The argument rests on establishing that reality is mathematical structure via an exhaustive pattern-randomness dichotomy, then showing that algorithmic probability provides the natural measure over structure-space. Bell correlations, rather than posing puzzles about nonlocality, become structural features of unified mathematical objects. The position explains why simpler physical theories tend to be true: they describe higher-measure structures.

Simple is probable. Many-Worlds is Occam’s Razor. Collapse is not.

1. Introduction

Physicists often prefer the Many-Worlds Interpretation on grounds of parsimony: it adds nothing to quantum mechanics beyond the Schrödinger equation. But what justifies this preference? Is simplicity merely aesthetic, or does it track something deeper?

I argue the preference is justified by measure theory. Under algorithmic probability—Solomonoff’s framework extended from epistemology to ontology—simpler descriptions correspond to higher-measure structures. MWI dominates because structures describable with minimal Kolmogorov complexity are more probable.

The argument proceeds in three steps. Section 2 establishes that reality is mathematical structure. Section 3 introduces the Simplicity Dominance Principle (SDP): the measure over structures is algorithmic probability. Section 4 applies SDP to quantum interpretation selection.

2. Reality as Mathematical Structure

2.1 The Dichotomy

All existence is either patterned or not-patterned (random). Both are mathematical.

Patterns are regularities expressible through mathematical relations. Randomness is characterized via algorithmic incompressibility—a string is random if and only if its Kolmogorov complexity K approximately equals its length. There is no third category. For something to exist determinately, it must have features distinguishing it from nonexistence. Those features are either patterned or random, and thus mathematical.

This is a logical partition, exhaustive in the way “A or not-A” is exhaustive.

2.2 From Isomorphism to Identity

Physical structures are isomorphic to mathematical structures by definition of “physical structure.” What distinguishes them? Not their relationships (identical by isomorphism). Not causal powers (supervening on relations). Not observables (reducing to relations).

The only candidate is non-relational substrate. But substrate without properties is indistinguishable from nothing. By the Identity of Indiscernibles, physical and mathematical structure are identical. Reality is mathematical structure, identical to it.

3. The Simplicity Dominance Principle

3.1 The Measure Problem

If reality is mathematical structure, all self-consistent structures exist. This raises the measure problem: among all structures, which should observers expect to inhabit? A clarification is needed. One can write down a Lagrangian (kinetic minus potential energy) with contradictory symmetry requirements, or assert that a particle simultaneously has definite position and momentum beyond the Heisenberg bound. The notation exists as a consistent string of symbols, carried by consistent physical media. But the physics it purports to describe cannot be instantiated as a functioning structure: the constraints are mutually exclusive under standard quantum mechanics. To make such a claim coherent requires modifying the framework, and that modification is itself an act of consistent reasoning. Inconsistent content can only be rescued by consistent mathematics. This is the sole building material.

Since reality *is* structure, the measure is discovered. We seek the natural measure over structure-space.

3.2 Algorithmic Probability

I propose:

$$\text{SDP: } P(S) = 2^{-K(S)}$$

where $K(S)$ is the Kolmogorov complexity of S 's generating rules.

Algorithmic probability is uniquely motivated:

Dominance: Solomonoff's completeness theorem establishes that algorithmic probability dominates all computable measures. No computable alternative systematically outperforms it.

Invariance: The measure is machine-independent up to multiplicative constants. Different universal Turing machines yield K values differing by at most an additive constant.

Coherence: SDP explains why Occam's Razor works. If simpler theories were not more likely true, parsimony's success in science would be miraculous.

Solomonoff intended algorithmic probability as epistemic: how to reason given uncertainty. SDP extends this to ontology: the measure governs existence as well as reasoning. We reason correctly about reality because algorithmic probability governs both inference and structure.

4. Application to Quantum Mechanics

4.1 Comparing Interpretations

Consider the major quantum interpretations by descriptive complexity:

Interpretation	What it adds to Schrödinger equation	Relative K
Many-Worlds	Nothing	Minimal
Copenhagen	Collapse postulate, measurement definition, classical/quantum cut	Higher
Bohmian	Hidden variables, pilot wave, guiding equation	Higher
GRW	Spontaneous collapse mechanism, two new constants	Higher

MWI takes the formalism literally: unitary evolution is complete. The wavefunction never collapses; all branches exist. This requires no additional postulates.

Why does each branch yield a definite value? A measurement couples the system to an apparatus through a specific observable. Observables are Hermitian operators (mathematical objects whose outputs are always real numbers), whose

eigenvalues (the specific values a measurement can yield) are the possible outcomes. Decoherence does the rest (Zurek, 2003). The apparatus entangles with the environment (air molecules, photons, thermal noise). States corresponding to different eigenvalues become effectively independent: they can no longer interfere. Each becomes its own branch with one definite eigenvalue. Decoherence is coarse-graining (averaging over fine details so many microstates map to one macrostate): the environment averages over quantum details. Many microstates, one macrostate per branch. The eigenvalue is what survives environmental averaging.

Collapse interpretations must specify: When does collapse occur? What counts as measurement? Where is the classical/quantum boundary? These are not answers within quantum mechanics but additions to it.

Hidden-variable theories add entire ontological layers: particles with definite positions guided by waves, nonlocal influences that somehow avoid signaling paradoxes.

Under SDP, the interpretation with minimal K dominates the measure. MWI wins.

A deeper argument reinforces this beyond parsimony. Under collapse interpretations, a specific measurement outcome is selected from the probability distribution. The generating rule (the wavefunction) has finite K . But the specific random outcome is algorithmically incompressible: it requires infinite specification complexity to determine why THIS value rather than another. There is a missing explanatory level between the finitely specifiable rule and the infinitely complex instantiation. Under MWI, this gap vanishes: all outcomes occur, no specific value requires selection, no infinite K appears at any level. MWI thus wins on explanatory completeness as well as parsimony. Discrete structures share this property: every element exists because a generating rule placed it, with no explanatory gap at any level.

4.2 Bell Correlations as Structure

Bell's theorem shows entangled particles correlate more strongly than any local hidden-variable theory permits. This is often framed as a puzzle about nonlocality.

Mathematical structural realism dissolves the puzzle. If reality is mathematical structure, and space is structure within mathematics (not the container of mathematics), then spatially separated particles need not "signal" across distance. They are one mathematical pattern described by a single entangled state vector.

Bell correlations are not spooky action at a distance. They are mathematics. The correlations are structural features of a unified mathematical object, not causal influences propagating through space.

Many-worlds treats branches as interacting. Matheism treats the whole wave-function as one mathematical structure; what looks like branches interacting is global reversible math viewed from inside irreversible functions, our forward arrow of time. The statue does not argue with itself. The photon is not in two places talking. One structure, multiple perspectives; the arithmetic of reality seen from inside any of its parts.

4.3 The Born Rule

Why do observed frequencies match $|\Psi|^2$? This remains an active research area. Deutsch (1999) and Wallace (2012) have argued that the Born rule follows from decision-theoretic rationality constraints within MWI: rational agents in branching universes must weight outcomes by squared amplitudes. This derivation does not require counting branches or assigning them intrinsic probabilities.

The derivation is contested. Kent (2010) argues that Deutsch-Wallace assumes a specific notion of rationality that is not forced by the physics. Albert (2010) objects that the derivation smuggles in probability assumptions it claims to derive. These objections have force, but they target the specific derivation strategy, not the compatibility of the Born rule with MWI. If an additional postulate is required to recover the Born rule, $K(\text{MWI})$ increases by the complexity of that postulate. Under SDP, this matters: MWI's parsimony advantage over other interpretations depends on the Born rule being derivable rather than postulated. The question is open. We note that even with an additional Born rule postulate, MWI's total K remains lower than collapse interpretations, which require both the Born rule AND an infinite- K specification of which outcome was selected at each measurement event.

The present framework is compatible with decision-theoretic approaches. SDP selects MWI as the dominant interpretation; within MWI, the Born rule emerges from rationality constraints (if Deutsch-Wallace succeeds) or from a compact additional postulate (if it does not). Either way, MWI avoids the infinite $K(\text{O})$ cost that collapse incurs. These are complementary, not competing claims.

A further SDP-compatible conjecture: high-amplitude branches may have lower K (less information needed to specify observer location within them), aligning algorithmic probability with Born probabilities at a deeper level. This remains plausible but unproven and is not required for the main argument.

4.4 Clarification

MWI is favored by SDP for structures like ours, but the Level IV multiverse contains all consistent structures. Structures where collapse is fundamental exist. Structures with genuinely nonlocal hidden variables exist. All consistent quantum interpretations correspond to structures that exist.

A note on superdeterminism, which rejects statistical independence between experimenter choices and hidden variables. The mainstream “extreme fine-tuning”

objection is a category error under algorithmic probability. It assumes initial-condition correlations are brute facts with high K . But if the correlations follow from a simple generating rule, K is low and the measure favors them. Genuinely random collapse outcomes have infinite K : every specific eigenvalue selection requires specification beyond the wave function. Both superdeterminism and MWI avoid this infinite- K cost. The fine-tuning objection inadvertently argues against collapse interpretations, not against superdeterminism. MWI retains a parsimony advantage (the Schrödinger equation alone requires no additional postulates) but this advantage holds only until a superdeterministic generating rule with lower total K is discovered. The competition is mathematical, not philosophical: whichever model has lower K wins. The “universal conspiracy” dissolves once correlations are recognized as outputs of compact generating functions.

MWI dominates the measure. It is not the only option that exists.

5. Objections

“MWI adds infinite branches-that’s complex!”

Branches are consequences, not axioms. Complexity is measured by description length, not by what the description entails. A short program can generate infinite output. The Schrödinger equation is simple; its solutions are vast.

“This is untestable metaphysics.”

The position makes predictions. If the final theory of physics has irreducibly high complexity, SDP is effectively falsified. More immediately: quantum computing provides indirect evidence for Many-Worlds. Superposition doing computational work suggests branches exist rather than being mere mathematical fiction.

“Why algorithmic probability?”

Algorithmic probability has unique mathematical properties: dominance over computable measures, invariance under machine choice. The critic asking “why this measure?” must produce an alternative satisfying comparable desiderata. None exists.

“The ontological question is misguided.”

Healey (2017) argues from a pragmatist position that quantum states guide agents without depicting reality, making the question “which interpretation is real?” ill-posed. Fuchs’s QBism (Fuchs, Mermin, and Schack, 2014) similarly treats quantum states as expressions of agent beliefs. Both approaches sidestep the interpretive competition by declining ontological commitment. The present argument engages this challenge indirectly: SDP is not a claim about which interpretation “really” describes the world. It is a claim about measure over mathematical structures. If all consistent structures exist (as PRD entails), then

structures with lower K have higher measure, and the question “which structure do we inhabit?” is answered by self-location within the measure-dominant class. Pragmatism and QBism are correct that agents need not settle the ontological question to use quantum mechanics successfully. But the measure question remains: among all structures containing agents who use quantum mechanics, which interpretive architecture dominates? SDP answers this without requiring the ontological commitment that pragmatism declines.

6. The Three-Way K -Comparison

SDP is not merely an argument for MWI. It is a principled framework for evaluating all quantum interpretations. The total specification complexity of a physical theory is $K(L) + K(I) + K(O)$:

Collapse interpretations: $K(L_{\text{collapse}})$ includes the Schrödinger equation plus the measurement postulate (Born rule and projection). $K(O)$ grows without bound: each measurement event selects one eigenvalue from the probability distribution, and that selection is additional information not determined by the theory. Over a universe’s history, $K(O)$ is effectively infinite, an algorithmically incompressible sequence of random outcomes.

Many-Worlds: $K(L_{\text{MWI}}) = K(\text{Schrödinger equation})$ alone. $K(O) = 0$: all outcomes occur, no selection needed. The cost is ontological (branching multiverse) but not complexity-theoretic (no additional specification).

Superdeterminism: $K(L_{\text{SD}}) + K(I_{\text{SD}})$, with $K(O) = 0$: outcomes follow from initial conditions. The fine-tuning objection claims $K(I_{\text{SD}})$ is large, but this assumes initial-condition correlations are brute facts. A full treatment of this category error, including engagement with six recent formulations of the objection, is given in Bernstein (2026b).

MWI currently leads. However, two unresolved problems may represent hidden K costs for MWI:

The Born rule. If the Deutsch-Wallace derivation fails (Kent, 2010; Albert, 2010), an additional postulate is required, increasing $K(L_{\text{MWI}})$. Even with this postulate, MWI avoids the unbounded $K(O)$ of collapse, but the gap between MWI and superdeterminism narrows.

The preferred basis. Decoherence selects the position basis in practice (Zurek, 2003; Wallace, 2012). If decoherence requires supplementary assumptions about the Hamiltonian structure or the initial state, $K(I)$ increases. This is a shared cost with superdeterminism: both depend on initial conditions, and the question is whose initial conditions have lower K .

The competition is mathematical. The framework decides it.

7. The Collapse Specification Gap

A deeper argument against collapse emerges from the distinction between generating rules and outputs. A short rule can generate a continuum distribution: quantum mechanics does exactly this, where the wavefunction is finitely specifiable and measurement outcomes follow the Born rule. The randomness does not add to specification complexity because it is drawn from the distribution, not specified point by point. But this saving works cleanly only under Many-Worlds, where all outcomes occur and no specific random value needs specification. Under collapse, the specific outcome at each measurement has no generating reason: infinite specification complexity hiding behind a short instruction. MWI is the only interpretation with no explanatory gap at the instantiation level. Collapse is not merely expensive under SDP. It requires the unconstructible: specific random outcomes with no generating rule. Permitting the unconstructible borders on square circles.

8. The Arrow of Computation Strengthens MWI

A further independent argument favors MWI from the structure of computation itself (Bernstein, 2026h). Nearly all computations are many-to-one functions: $1+1=2$ is deterministic, but $2=x+y$ has infinitely many solutions. The inverse function does not exist: not hard to compute, but nonexistent. This is a structural fact about mathematics, independent of physics.

Reversible computing (Bennett, 1973) does not change this. It computes a different bijective function that stores inputs alongside outputs. The original function's inverse still does not exist. Each timeline lacks the backward path, even if the multiverse globally stores all paths. The arrow is absolute.

Under collapse interpretations, the wavefunction itself is many-to-one: superposition collapses to one outcome, destroying information at the substrate level. The arrow exists at both the function level and the substrate level. Under MWI, the substrate remains unitary (one-to-one). The arrow exists only at the function level, where it is a structural fact requiring no additional postulate.

MWI thus provides the cleanest architecture: reversible substrate plus irreversible functions. The arrow of time needs no thermodynamic explanation and no collapse mechanism. It follows from the mathematics of many-to-one functions operating within a unitary substrate. Collapse interpretations are overkill: they break reversibility at the substrate level when the arrow already exists at the function level. Under SDP, overkill means higher K and lower measure.

9. Conclusion

Quantum interpretation selection is not a philosophical dispute to be settled by taste. Under algorithmic probability extended to ontology, the total specification complexity $K(L) + K(I) + K(O)$ determines which interpretations dominate the measure.

Collapse interpretations incur unbounded $K(O)$, the only provably growing complexity cost. This alone places them below both deterministic alternatives. MWI currently leads on parsimony: the Schrödinger equation alone, with no additions, has minimal K among known frameworks. Superdeterminism is rehabilitated once the fine-tuning objection is recognized as a category error, specificity is not complexity.

The final ranking depends on model-specific K computations that are not yet available. What SDP provides is the framework for making those computations meaningful: not which interpretation we prefer, but which interpretation the mathematics prefers.

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